

Notes on Regular Epimorphisms and the Monadicity of the Pullback Functor in an Exact Category

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These notes are based on the work of Sobral and Tholen [ST92], with some ideas drawn from Janelidze and Tholen [JT91].

Let $\mathcal{A}$ be a category with pullbacks and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. Consider the monad $T = p^*p_!$ associated to p . Let (C, γ) be an object in $(C \downarrow E)$ and let $(C, \gamma; \xi)$ be an object in $(\mathcal{A} \downarrow E)^T$ with $\xi : T(C) \rightarrow C$ an algebra.

Consider the pullback in $\mathcal{A}$.

$$\begin{array}{ccc} E \times_B C & \xrightarrow{\text{proj}_2} & C \\ \text{proj}_1 \downarrow & \searrow \gamma & \downarrow p\gamma \\ E & \xrightarrow{p} & B \end{array}$$

Thus, we have the span in $\mathcal{A}$:

$$\begin{array}{ccc} & E \times_B C & \\ \text{proj}_2 \swarrow & & \searrow \xi \\ C & & C \end{array}$$

Since ξ is also a morphism in $(\mathcal{A} \downarrow E)$, we have that $\gamma\xi = \text{proj}_1$

$$\begin{array}{ccc} E \times_B C & \xrightarrow{\xi} & C \\ & \searrow \text{proj}_1 & \swarrow \gamma \\ & E & \end{array}$$

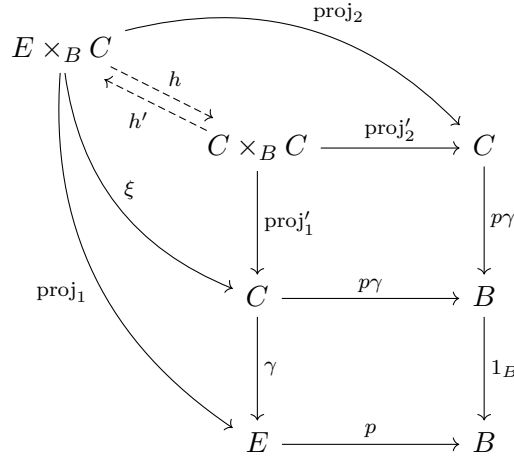
Proposition 1. *Let $\mathcal{A}$ be a finitely complete category and let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. For any object $(C, \gamma; \xi)$ in $(\mathcal{A} \downarrow E)^T$, with T the monad associated to p . The pair (proj_2, ξ) is an equivalence relation in $\mathcal{A}$*

Proof. First we show that (proj_2, ξ) is a regular epi is an internal relation. So we need

to show that $\langle \text{proj}_2, \xi \rangle$ is a mono. Consider the following diagram of solid arrows¹. It commutes, in fact

- $1_B(p\gamma)\text{proj}_2 = p\text{proj}_1 = p\gamma\xi$
- $1_B(p\gamma)\text{proj}'_2 = 1_B(p\gamma)\text{proj}'_1 = p\gamma\text{proj}'_1$

Thus we can consider the morphism h' induced by the universal property of the pullback from proj'_2 and $\gamma\text{proj}'_1$ and h similarly induced from proj_2 and ξ .



In order to prove that $\langle \text{proj}_2, \xi \rangle$ is a mono, we need to show that h is a mono. Since $(\text{proj}_1, \text{proj}_2)$ is a jointly monic pair, by the following compositions

- $\text{proj}_1 h' h = \gamma \text{proj}'_1 h = \gamma \xi = \text{proj}_1 = \text{proj}_1 1_{E \times_B C}$
- $\text{proj}_2 h' h = \text{proj}'_2 h = \text{proj}_2 = \text{proj}_2 1_{E \times_B C}$

we have that $h' h = 1_{E \times_B C}$. This implies that h is a mono, in fact suppose that f and g are two morphisms such that $hf = hg$, then we have

$$\begin{aligned}
 hf &= hg \\
 \implies h' hf &= h' hg \\
 \implies 1_{E \times_B C} f &= 1_{E \times_B C} g \\
 \implies f &= g
 \end{aligned}$$

¹[ST92] assumes the existence of coequalizers, which is not the case in our setting. However, this is not a problem: at this point, we can use the morphism $p\gamma$ instead of the coequalizer of (proj_2, ξ) , as it serves our purpose. Then, *a posteriori*, this proposition ensures, under the assumption of exact category, that the pair (proj_2, ξ) has a coequalizer, which we will make use of.

Consider now the canonical morphism $i : C \times_B C \rightarrow C \times C$ induced by the universal property of the product. We have that i is a mono, in fact f and g are two morphisms such that $if = ig$, then we have

$$\begin{array}{ccccc}
 & & & C & \\
 & & \text{proj}'_1 \nearrow & \downarrow \pi_1 & \\
 A & \xrightarrow[f]{g} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & \text{proj}'_1 \searrow & \uparrow \pi_2 & \\
 & & & C &
 \end{array}$$

- $if = ig \implies \pi_1 if = \pi_1 ig \implies \text{proj}'_1 f = \text{proj}'_1 g$
- $if = ig \implies \pi_2 if = \pi_2 ig \implies \text{proj}'_2 f = \text{proj}'_2 g$

Since $(\text{proj}'_1, \text{proj}'_2)$ is a jointly monic pair, we have that $f = g$. Thus, i is a mono. Since i and h are both monos, we have that ih is a mono.

By the following commutative diagram

$$\begin{array}{ccccc}
 & & \text{proj}_1 \nearrow & & \\
 & & & C & \\
 & & \text{proj}'_1 \nearrow & \downarrow \pi_1 & \\
 E \times_B C & \xrightarrow{h} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & \text{proj}'_2 \searrow & \uparrow \pi_2 & \\
 & & & C & \\
 & & \text{proj}_2 \searrow & &
 \end{array}$$

we conclude that $ih = \langle \text{proj}_2, \xi \rangle$, so $\langle \text{proj}_2, \xi \rangle$ is a mono.

We proved that (proj_2, ξ) is an internal relation, now we need to show that it is an equivalence, ie it is reflexive, symmetric and transitive.

- Claim: (proj_2, ξ) is reflexive.

Called η the unit of the monad, we have that $(\text{proj}_2, \xi)\eta_{(C, \gamma)} = \langle 1_C, 1_C \rangle$. In fact $\xi\eta_{(C, \gamma)} = 1_C$ and by the definition of the unit of the monad, we have that $\text{proj}_2\eta_{(C, \gamma)} = \text{proj}_2\langle \gamma, 1_C \rangle = 1_C$.

$$\begin{array}{ccccc}
 & C & & & \\
 & \swarrow 1_C & \downarrow \eta_{(C, \gamma)} & \searrow 1_C & \\
 C & \xleftarrow{\text{proj}_2} & E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

- Claim: (proj_2, ξ) is symmetric.

Consider the following diagram, where proj_1'' and proj_2'' are the projections of $E \times_B C$. Let l be the morphism induced by the universal property of the pullback from $1_{E \times_B C}$ and γproj_2 , in particular it is the unique such that

$$\gamma \text{proj}_2 = l \text{proj}_1'' = l \text{proj}_1 (1_E \times \text{proj}_2)$$

and

$$\begin{array}{ccccc}
 & & 1_{E \times_B C} = l \text{proj}_2'' & & \\
 & & \text{---} 1_{E \times_B C} \text{---} & & \\
 E \times_B C & \xrightarrow{\quad l \quad} & E \times_B (E \times_B C) & \xrightarrow{\quad \text{proj}_2'' \quad} & E \times_B C \\
 \downarrow \text{proj}_2 & & \downarrow 1_E \times_B \text{proj}_2 & & \downarrow \text{proj}_2 \\
 & & E \times_B C & \xrightarrow{\quad \text{proj}_2 \quad} & C \\
 & & \downarrow \text{proj}_1 & & \downarrow p\gamma \\
 C & \xrightarrow{\quad \gamma \quad} & E & \xrightarrow{\quad p \quad} & B
 \end{array}$$

Define $\sigma = (1_E \times_B \xi)l$. Observe that

$$\text{proj}_2 \sigma = \text{proj}_2 (1_E \times_B \xi)l = \xi 1_{E \times_B C} = \xi$$

It remains to show that $\xi \sigma = \text{proj}_2$. This holds because of the following compositions and the fact that $(\text{proj}_1, \text{proj}_2)$ is a jointly monic pair.

- $\text{proj}_1 (1_E \times_B \text{proj}_2)l = \text{proj}_1'' l = \gamma \text{proj}_2$
- $\text{proj}_1 \eta_{(C, \gamma)} \text{proj}_2 = \text{proj}_1 \langle \gamma, 1_C \rangle \text{proj}_2 = \gamma \text{proj}_2$
- $\text{proj}_2 (1_E \times_B \text{proj}_2)l = \text{proj}_2 \text{proj}_2'' l = \text{proj}_2 1_{E \times_B C} = \text{proj}_2$
- $\text{proj}_2 \eta_{(C, \gamma)} \text{proj}_2 = \text{proj}_2 \langle \gamma, 1_C \rangle \text{proj}_2 = 1_C \text{proj}_2 = \text{proj}_2$

$$\begin{array}{ccccc}
 & & E \times_B C & & \\
 & \swarrow \xi & \downarrow \sigma = (1 \times_B \xi)l & \searrow \text{proj}_2 & \\
 C & & E \times_B C & & C \\
 & \swarrow \text{proj}_2 & & \searrow \xi & \\
 & & E \times_B C & &
 \end{array}$$

- Claim: (proj_2, ξ) is transitive.
Consider the pullback (???)

$$\begin{array}{ccc} (E \times_B C) \times_C (E \times_B C) & \longrightarrow & E \times_B C \\ \downarrow & & \downarrow \xi \\ E \times_B C & \xrightarrow{\text{proj}_2} & C \end{array}$$

□

In an exact category each equivalence relation is a kernel pair and by $(-)$ has a coequalizer. Thus in an exact category (proj_2, ξ) has a coequalizer. Let $q : C \rightarrow Q$ be such coequalizer. Observe that considering proj_2 and ξ as morphism in $(\mathcal{A} \downarrow B)$, then q is a morphism in $(\mathcal{A} \downarrow B)$ as well by the universal property. So, called δ the morphism induced by the universal property of the coequalizer from proj_2 and ξ , we have that q with codomain (Q, δ) is also the coequalizer of the pair in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccccc} E \times_B C & \xrightarrow{\xi} & C & \xrightarrow{q} & Q \\ & \searrow \text{proj}_2 & \downarrow p\gamma & \swarrow \delta & \\ & & B & & \end{array}$$

$p\text{proj}_1$ (arrow from $E \times_B C$ to B)

Consider the comparison functor $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$ defined as $K(D, \zeta) = (p^*(D, \zeta), p^*(\epsilon_{(D, \zeta)}))$, where ϵ is the counit of the adjunction, defining $\epsilon_{(D, \zeta)}$ to be the projection from $E \times_B D$ to D of the pullback of p along ζ .

In our context of exact category, turns out that the comparison functor K has a left adjoint $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$ defined as $L(C, \gamma; \xi) = (Q, \delta)$, where (Q, δ) is the codomain of coequalizer of the pair (proj_2, ξ) in $(\mathcal{A} \downarrow B)$.

Let us introduce the counit ρ and counit α of the adjunction $K \dashv L$.

For each object (D, ζ) in $(\mathcal{A} \downarrow B)$, the counit is a morphism of the kind $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$. First we compute $K(D, \zeta)$.

$$K(D, \zeta) = (E \times_B D, \pi_1; p^*(\pi_2))$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\pi_2} & D \\ \downarrow \pi_1 & & \downarrow \zeta \\ E & \xrightarrow{p} & B \end{array}$$

Then, call $q : E \times_B D \rightarrow Q$ the coequalizer of the equivalence relation $(\text{proj}_2, p^*(\pi_2))$, with codomain (Q, δ) and compute $L(K(D, \zeta))$. We define in the diagram below the morphism

$\rho_{(D,\zeta)}$ by the universal property of the coequalizer.

$$\begin{array}{ccccc}
E \times_B (E \times_B D) & \xrightarrow[p^*(\pi_2)]{\text{proj}_2} & E \times_B D & \xrightarrow{q} & Q \\
\downarrow \text{proj}_2 & & \downarrow \pi_2 & \nearrow \rho_{(D,\zeta)} & \\
E \times_B D & \xrightarrow{\pi_2} & D & & \\
\downarrow \pi_1 & & \downarrow \zeta & \nearrow \delta & \\
E & \xrightarrow{p} & B & &
\end{array}$$

Observe that δ was also defined as the morphism induced by the universal property of the coequalizer, considering the object $E \times_B D$ as an object in $(\mathcal{A} \downarrow B)$ via $p\pi_1$. Thus, by the commutation of the diagram, $\rho_{(D,\zeta)}$ is a morphism in $(\mathcal{A} \downarrow B)$.

(UNIT TO DO. Why are the two definitions of the unit α the same? Does ξ coequalize $(p^*(\xi), p^*(\text{proj}_2))$))

Our goal is to show that p^* is a monadic functor, ie that the comparison functor K is an equivalence of categories. In order to do so, we aim to prove that the unit α and the counit ρ of the adjunction $K \dashv L$ are pointwise isomorphisms.

Proposition 2. *In an exact category if $p : E \rightarrow B$ is a regular epi, then the counit ρ of the adjunction $K \dashv L$ is a pointwise isomorphism.*

Proof. We aim to prove that for each object (D, ζ) in $(\mathcal{A} \downarrow B)$, the morphism $\rho_{(D,\zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ is an isomorphism. Consider the diagram of the definition of the counit $\rho_{(D,\zeta)}$. An exact category is also regular, in particular regular epi are pullback stable. If p is a regular epi, then π_2 is a regular epi as well. The top left square is a pullback by (CITE), thus $(p^*(\pi_2), \text{proj}_2)$ is the kernel pair of π_2 . Since π_2 is a regular epi, then, by (CITE), it is the coequalizer of its kernel pair $(p^*(\pi_2), \text{proj}_2)$. We have that q and π_2 are coequalizers of the same pair, thus the morphism $\rho_{(D,\zeta)}$ defined by the universal property of the coequalizer q is actually an isomorphism. $\square$

References

- [JT91] G. Janelidze and W. Tholen. “How algebraic is the change-of-base functor?” In: *Category Theory, proceedings of the International Conference held in Como, Italy, July 22-28, 1990*. Lecture Notes in Mathematics 1488. Berlin and Heidelberg: Springer, 1991, pp. 174–186.
- [ST92] M. Sobral and W. Tholen. “Effective descent morphisms and effective equivalence relations”. In: *Category Theory 1991, proceedings of an International Summer Category Theory Meeting held June 23-30, 1991*. Canadian Mathematical Society Conference Proceedings 13. Providence, RI: American Mathematical Society, 1992, pp. 421–433.